Churn Risk Modeling

Leakage-aware classification and monthly-charge modeling

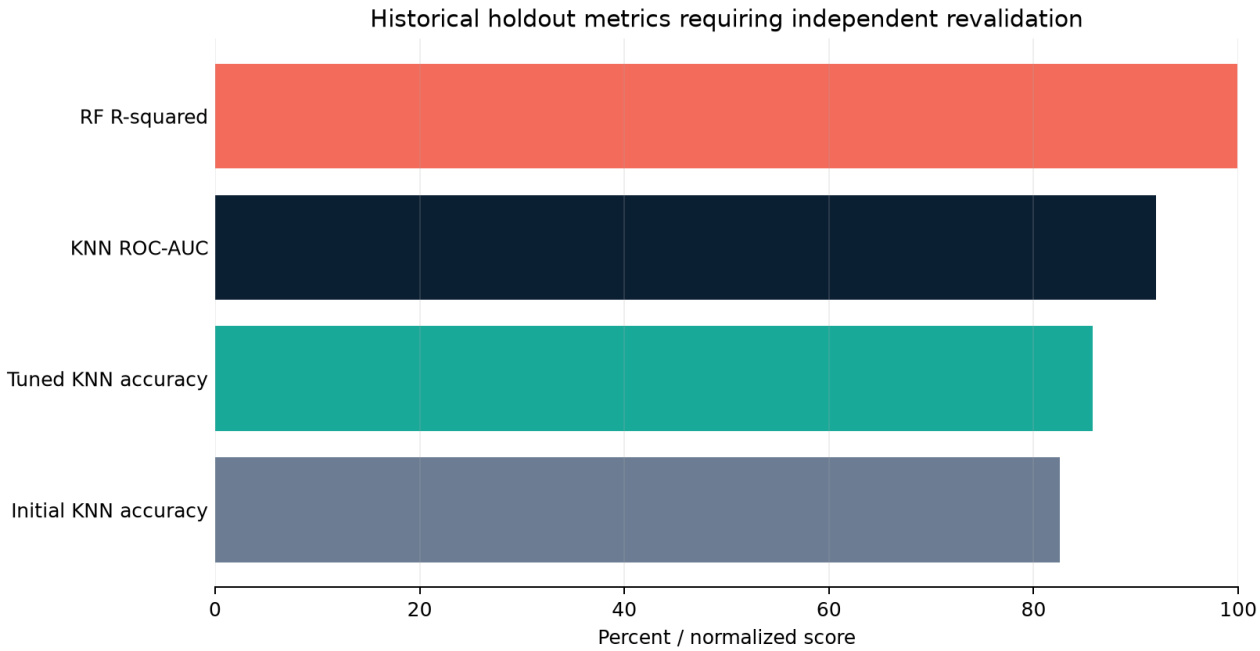
Portfolio Case Study

Executive Summary

This case study combines churn classification with monthly-charge regression. The historical KNN experiment improved from 82.6% to 85.8% accuracy after tuning and reached 0.92 ROC-AUC. The historical random-forest regression reported MSE of 0.08 and R-squared of 0.999; because that performance is unusually strong, the public implementation emphasizes leakage-safe preprocessing and independent holdout validation before business use.

Decision Question

Can supervised learning identify churn risk and model monthly charges while keeping preprocessing inside reusable pipelines?



Methods

- Split data before fitting transformations and kept preprocessing inside scikit-learn pipelines.
- Standardized numeric variables and one-hot encoded categorical variables.
- Evaluated K-nearest-neighbor classification with accuracy and ROC-AUC.
- Evaluated random-forest regression with MAE, RMSE, and R-squared on held-out observations.

Evidence and Findings

Evidence	Observed result
KNN classification	Historical tuned performance reached 85.8% accuracy and 0.92 ROC-AUC.
Model improvement	Hyperparameter tuning increased historical accuracy by 3.2 percentage points.
Charge regression	Historical results reported MSE of 0.08 and R-squared of 0.999.
Public implementation	Reusable pipelines isolate preprocessing from evaluation to reduce leakage risk.

Limitations

- The historical regression result is strong enough to require explicit leakage and target-proxy checks.
- Accuracy can hide class imbalance and does not determine the business value of interventions.
- The public repository excludes the restricted source dataset and cannot reproduce historical scores without an authorized equivalent input.

Recommended Next Actions

- Revalidate against an untouched temporal holdout before using the historical performance claims operationally.
- Choose classification thresholds using intervention capacity, recall, and expected value.
- Audit monthly-charge predictors for post-outcome fields and direct pricing proxies.

Reproducibility

The companion repository contains typed Python modules, tests, dependency declarations, data-access notes, and a rendered analysis notebook. Restricted source data and the working notebook are intentionally excluded from the public repository.

Results are reported as historical analytical evidence. They should be reproduced on an authorized equivalent dataset before operational use.